



Ped-ID interesting case conference

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27th February 2024



Identification data

- Case preterm female NB, GA 36+6 weeks, BBW 2,380 gm
- Place of birth: McCormick hospital
- Date of birth: 30th January 2024
- Consult ped ID: 22/2/2024



Maternal history

- Maternal age 37 years, G4P2012, GA 36+6 wks
- No underlying disease, ANC at private hospital
- Blood group O, Rh negative
- VDRL NR, Anti-HIV NR/NR, HBsAg negative
- Lab Hb 12.9/11.3, Hct 38/35%, MCV 90 fL
- Thalassemia screening: not couple at risk
- NIPT test: 46, XX
- U/S anomaly at GA 20+5 weeks
 - No gross anomaly seen
 - **Single umbilical artery**



Maternal history

- **Maternal Rh negative**
 - Husband: Blood group A, Rh positive
 - W/U Coomb's test at GA 14 week → DAT/IAT negative
 - W/U anti-D at GA 29 week → negative
 - S/P Rhogam at GA 29+2 week



Natal history

- C/S due to previous C/S at McCormick hospital
- Apgar 2-7-9, PPV 2 cycle, suction clear secretion then transfer to NS
- On O2 via face mask 10 LPM → SpO2 95-96%
- **Imp:** Respiratory distress, R/O Early onset neonatal sepsis
- **Management:** NPO, On O2 cannula 2 LPM, Start **ampicillin 50 MKDose IV q 8 hr, Gentamicin 4 MKDose IV q 24 hr**
- **Progression:**
 - หลังรับ feed มีอาเจียนเป็นน้ำดีปน old blood 1 รอบ อุจจาระ meconium ไม่มีมูกเลือดปน
 - Abdomen: soft, not tender, normoactive bowel sound
- **Management:**
 - NPO, retained OG tube (Bile content)
 - IV fluid
 - CXR include abdomen: hyperaeration with asymmetrical opacities at both lungs, air in the stomach
- **Impression:** **Congenital pneumonia with R/O partial bowel obstruction (midgut volvulus)**
- Refer CMU for R/O midgut volvulus



Post natal history

- **History of congenital pneumonia**

- Sepsis screening at birth (McCormick hospital) = 0
 - ANC 15,528, mESR 1 mm/hr, CRP <6 g/dL
- Sepsis screening at 12-24 hr = 0
 - ANC 13310, mESR 2 mm/hr, CRP < 2 g/dL, I:T 0.12
 - H/C (31/1/67):
 - I: No growth in 5 days* I
 - II: **S. epidermidis*I (contaminate)**
→ repeat H/C (1/2/67): no growth in 5 days * II
- Management:
 - **Continue Ampicillin + Gentamicin IV total 7 days (DOL 1-7)**
 - On O2 cannula 2 LPM



#	Qty	Code	Organism
▶ 1		STPEP	Staphylococcus epidermidis
		</	



Post natal history

- **R/O midgut volvulus**
- **Management:**
 - NPO total 3 Days (30/1/67 - 2/2/67)
 - Request U/S whole abdomen emergency (31/1/67)
 - Upper GI study



U/S whole abdomen (DOL1, 31/1/67)

- R/O midgut volvulus

EMERGENCY ULTRASOUND OF THE WHOLE ABDOMEN

HISTORY: Case of near-term female GA 36+5 weeks with acute vomiting (bile and old blood). Void and pass meconium within first 24 hours. To evaluate midgut volvulus.

FINDINGS:

- Normal SMA and SMV relationship. No evidence of midgut volvulus or malrotation.
- No dilatation of the visualized bowel loops.
- Normal size and echogenicity of liver parenchyma. No focal mass is seen. The portal and hepatic veins are patent. No bile duct dilatation is detected.
- The gallbladder, spleen and partial visualized pancreas are unremarkable.
- Small size of bilateral kidneys. The right and left kidney lengths are 3.4 cm and 3.4 cm, respectively. Normal echogenicity of bilateral renal parenchyma. No dilatation of bilateral pelvicalyceal systems.
- A large cystic lesion in pelvic cavity with multilayered wall, could be markedly distended urinary bladder or duplication cyst.
- No ascites.

IMPRESSION:

- No evidence of midgut volvulus or malrotation.
- A large cystic lesion in pelvic cavity with multilayered wall, could be markedly distended urinary bladder or duplication cyst. Ultrasound after urinary catheter insertion is suggested.



Upper GI study (DOL2, 1/2/67)

EMERGENCY UPPER GI STUDY:

CONTRAST: Barium suspension feeding via orogastric tube, total volume of 30 mL.

HISTORY: Near-term female newborn with acute vomiting (bile and old blood). The child passed void and meconium within first 24 hours. Abdominal sonography showed no evidence of midgut volvulus. This study was done to evaluate for the cause.

FINDINGS:

- Abnormal position of the duodenal-jejunal (DJ) junction, located anterior to the duodenal bulb with abnormal course of the duodenal C-loop, located superior to the duodenal bulb
- Stenosis of the 3rd-4th parts of duodenal resulting in dilatation of stomach and more proximal duodenum, and delayed contrast passage into the more distal bowel loops/
- Unremarkable partially-opacified proximal jejunum
- No corkscrew appearance.
- No contrast leakage.

IMPRESSION:

- Midgut malrotation resulting in partial duodenal obstruction, no evidence of volvulus

- **Impression:** Midgut malrotation resulting in partial duodenal obstruction, no evidence of volvulus



Ped sx plan

- Evaluate เรื่อง midgut malrotation จาก clinical and film X-ray
- ➔ เนื่องจาก UGIS ตำแหน่ง Duodenum อยู่ทางด้านซ้ายและสูงเกินกว่าตำแหน่งที่ common ของ midgut malrotation และตำแหน่งของ IC Valve อยู่ทางด้านขวา
- ➔ **Less-likely midgut malrotation**
- **Impression: Feeding intolerance due to 1. ileus 2. pylorospasm**
- **Management:** start trophic feeding and step feeding ➔ มีปัญหาเรื่อง feeding intolerance อีกครั้ง content เหลือเป็น bile ➔ **Ped Sx suspected from 1. midgut malrotation 2. full bladder from neurogenic bladder**
- **Management:**
 - Retained foley catheter for release residual volume from bladder
 - Step feeding หลังจาก retained foley's catheter ➔ หลังจากนั้นผู้ป่วยยังคงมี feeding intolerance อยู่
- **Impression: Midgut malrotation**
- **Management:**
 - Set OR for EL to correct intestinal rotation divide ladd band with appendectomy (14/2/67)

CBC (DOL 1, 31/1/67)

CBC ...

☒ Hb	17.9 g/dL (12.0 - 16.0) (<7.0 >22.0) (High)
☒ Hct	56.1 % (36.0 - 48.0) (<21.0 >66.0) (High)
☒ WBC	19460 cells/cu.mm. (5000 - 10000) (High)

Differential Per 100 Cells ...

☒ Neutrophil (%)	68.4 % (40.0 - 74.0)
☒ Eosinophil (%)	3.0 % (0.0 - 7.0)
☒ Basophil (%)	0.6 % (0.0 - 2.0)
☒ Lymphocyte (%)	18.6 % (19.0 - 48.0) (Low)
☒ Monocyte (%)	9.4 % (3.0 - 9.0) (High)
☒ Platelet per cu.mm.	340000 cells/cu.mm. (140000 - 450000) (<30000 >1000000)
☒ RBC	5.84 x1000000 cells/cu.mm. (4.00 - 5.50) (High)
☒ MCV	96.1 fl. (80.0 - 99.0)
☒ MCH	30.7 pg. (27.0 - 31.0)
☒ MCHC	31.9 g/dL
☒ MPV	9.2 fL. (7.2 - 11.1)
☒ PDW	9.7 fL. (9.6 - 15.2)

Comment:-

☒ Absolute Neutrophil Count	13310 cells/cu.mm.
☒ Absolute Lymphocyte Count	3620 cells/cu.mm.
☒ NRBC	4.9 /100 WBC (<=0.0) (High)
☒ RDW	17.6 % (11.5 - 14.5) (High)
☒ Platelet large cell ratio	19.1 % (19.7 - 42.4) (Low)
☒ Absolute Neutrophil Count	13310 cells/cu.mm.
☒ Absolute Lymphocyte Count	3620 cells/cu.mm.

LABORATORY MEMBER



New onset sepsis (DOL13, 12/2/67)

- ใช้ BT 37.9 C°, 38.5 C° แหะนม ไม่เหนื่อย
- V/S: BT 38.5 C, HR 160 bpm, RR 40-60/min, SpO2 98% (RA)
- Lungs: clear and equal BS BL, no retraction
- Heart: regular rhythm, no murmur
- Abdomen: soft, not tender, OG content นมย่อย
- Ext: pulse 2+ all extremities, CRT < 2 sec
- **Impression:** Late onset neonatal sepsis
- **Management:**
 - Septic W/U, H/C, CRP
 - Set LP
 - CXR include abdomen
 - Empirical antibiotics: ceftazidime (150 MKDay) IV q 8 hr, Amikin (15 MKDay) IV q 12 hr

+ + Septic work up (12/2/67)

CBC	12/2/67
Hb (g/dL)	16.3
Hct (%)	49.7
WBC (cells/cu.mm)	14,960
Neutrophil(%)	58.3
Eosinophil(%)	5.3
Basophil (%)	0.5
Monocyte (%)	10.8
Lymphocyte(%)	25.1
Platelet	424,000

UA	12/2/67
WBC	30-50 HPF
RBC	3-5 HPF
Epithelium	0-1 HPF
Leukocyte esterase	3+
Bacteria	numerous
WBC clumping	positive
pH	7.0
Spec	1.007
Blood	2+

- CRP 26 mg/L
- Urine G/S: PMN 0-1 cell/L, GNB 0-5 cells, GPC 0-1 cell



CXR portable include abdomen (12/2/67)





Investigations (12/2/67)

- CSF profiles:
 - CSF **WBC 5, RBC 640 cells/cu.mm**, PMN 0%, MN 0%
 - CSF glucose 47 mg/dL (40-70), CSF protein 49 mg/dL (15-45)
 - CSF G/S: no bacteria found
 - CSF C/S: no growth in 2 days
- Hemoculture:
 - H/C (12/2/67) → report **GNB from peripheral vein * II (TTP 24 hr)**
 - Repeat H/C (13/2/67)
- Ped sx plan: set OR for EL to correct intestinal rotation with divided ladd band with appendectomy (14/2/67)



Operation (14/2/67)

- **Pre-op diagnosis:** Midgut volvulus with partial bowel obstruction
- **Operation:** EL for modified Ladd's procedure, diamond shape duodenoduodenostomy with Meckel's diverticulectomy with appendectomy
- **Post-op diagnosis:** Duodenal atresia from inferior mesenteric vein with annular pancreas (14/2/67)
- **Post-op management**
 - Ceftazidime (50 MKDose) IV q 8 hr
 - Amikin 15 MKDay IV q 24 hr
 - Add metronidazole (7.5 MKDose) IV q 8 hr



Hemoculture (12/2/67) → report 15/2/67

Lab L101823 Bacteria Hemoculture				
#	Qty	Code	Organism	
1		BKDCP	Burkholderia cepacia complex	
#	Code	Antibiotic	MIC	Result
1	CZ	Ceftazidime	2	S
2	LX	Levofloxacin	4	I
3	ME	Meropenem	4	S
4	ST	Trimethoprim-sulfamethoxazole	<=1	S

Bottle1#
(มีสีขาว)

Lab L101823 Bacteria Hemoculture				
#	Qty	Code	Organism	
1		BKDCP	Burkholderia cepacia complex	
#	Code	Antibiotic	MIC	Result
1	CZ	Ceftazidime	2	S
2	LX	Levofloxacin	4	I
3	ME	Meropenem	4	S
4	ST	Trimethoprim-sulfamethoxazole	<=1	S

Bottle2#
(ไม่มีสีขาว)



Urine culture (12/2/67) → report 16/2/67

#	Qty	Code	Organism
1	>=100,000 cfu	BKDCN	Burkholderia cenocepacia
2	10,000-100,000 cfu	ETCFL	Enterococcus faecalis

#	Code	Antibiotic	MIC	Result
1	CZ	Ceftazidime	2	S
2	LX	Levofloxacin	4	I
3	ME	Meropenem	2	S
4	ST	Trimethoprim-sulfamethoxazole	<=1	S

#	Qty	Code	Organism
1	>=100,000 cfu	BKDCN	Burkholderia cenocepacia
2	10,000-100,000 cfu	ETCFL	Enterococcus faecalis

#	Code	Antibiotic	MIC	Result
1	AM	Ampicillin	1	S
2	CI	Ciprofloxacin	1	S
3	G(500)	Gentamicin High Level (500)		R
4	LX	Levofloxacin	2	S
5	LNZ	Linezolid	1	S
6	P	Penicillin	4	S
7	VA	Vancomycin	1	S

Timeline

- Imp: **Late onset neonatal sepsis**
- Mx: LP, septic W/U
Report (16/2/67)
UTI from *B. cenocepacia* and *E. faecalis*
B. Cepacia complex septicemia

Dx. Congenital pneumonia

DOL1

DOL7

12/2/67

H/C: no growth in
5 days

13/2/67

14/2/67

17/2/67

Duodenal atresia S/P
duodenoduodenostomy

New onset late onset
neonatal sepsis

Ampicillin IV * 7 days

Gentamicin IV * 7 days

Ceftazidime (150 MKDay) IV q 8 hr

Day5

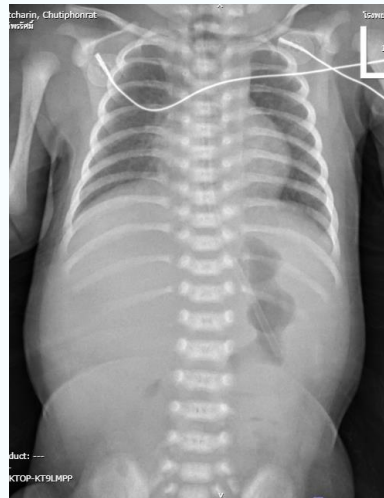
Amikin (15 MKDay) IV q 12 hr* 3 days

Metronidazole (22.5 MKDay) IV q 8 hr



Progression, post-op day 3 (17/2/67)

- Notify หายใจเหนื่อยมากขึ้น มีไข้
- V/S: **BT 37.9 C**, **RR 80/min**, **tachycardia**, no desaturation
- GA: nasal flaring
- Lungs: clear and equal BS BL, no grunting, **marked subcostal retraction**
- Heart: regular, no murmur
- Ext: **CRT 3 sec**
- **Impression:** **Late onset neonatal sepsis**
- **Management:**
 - On HHHFNC 3 LPM, FiO2 0.3
 - NSS 25 ml IV load in 15 min
 - Septic W/U
 - Set LP
 - Off ceftazidime → **Meropenem 120 MKDay IV q 8 hr**, **Vancomycin 40 MKDay IV q 8 hr**



Investigations (17/2/67)

CBC	17/2/67
Hb (g/dL)	12.6
Hct (%)	39.7
WBC (cells/cu.mm)	4,440
Neutrophil(%)	69.4
Eosinophil(%)	2.9
Basophil (%)	0.2
Monocyte (%)	2.0
Lymphocyte(%)	25.5
Platelet	224,000

- CSF profile
 - CSF glucose 50 mg/dL, protein 58 mg/dL
 - WBC 2 cells/cc,mm, RBC 0, MN 100%
 - CSF gram stain: no bacteria found
 - CSF C/S: no growth in 2 day
- CRP 111 mg/L



Investigations (17/2/67)

UA	17/2/67
WBC	30-50 HPF
RBC	10-20 HPF
Epi	none HPF
Leukocyte esterase	2+
Bacteria	rare
pH	6.0
Spec	1.010
Blood	2+

- **Urine gram stain (single catheter):**
 - no bacteria found
- **Urine culture (17/2/67): pending**



Hemoculture (17/2/67)

- H/C from peripheral vein (17/2/67): GNB (TTP 10 hrs)* I
- H/C from peripheral vein (17/2/67): GNB* I

+ Urine C/S (single cath, 17/2/67) → report 24/2/67

1	10,000-100,000 cfu	BKDC	Burkholderia cenocepacia	
#	Code	Antibiotic	MIC	Result
1	CZ	Ceftazidime	8	S
2	LX	Levofloxacin	4	I
3	ME	Meropenem	8	I
4	ST	Trimethoprim-sulfamethoxazole	<=1	S

- Repeat UA, urine G/S, urine culture (20/2/67)



Hemoculture (17/2/67) → report 21/2/67

#	Qty	Code	Organism
1		BKDCP	Burkholderia cepacia complex

#	Code	Antibiotic	MIC	Result
1	CZ	Ceftazidime	8	S
2	LX	Levofloxacin	8	R
3	ME	Meropenem	8	I
4	ST	Trimethoprim-sulfamethoxazole	<=1	S

Request

Bottle1#
(แก้วที่ 1)

Comment

Preliminary report : gram negative bacilli

Time to positive = 20 hours. 16 mins.

#	Qty	Code	Organism
1		BKDCP	Burkholderia cepacia complex

#	Code	Antibiotic	MIC	Result
1	CZ	Ceftazidime	8	S
2	LX	Levofloxacin	8	R
3	ME	Meropenem	8	I
4	ST	Trimethoprim-sulfamethoxazole	<=1	S

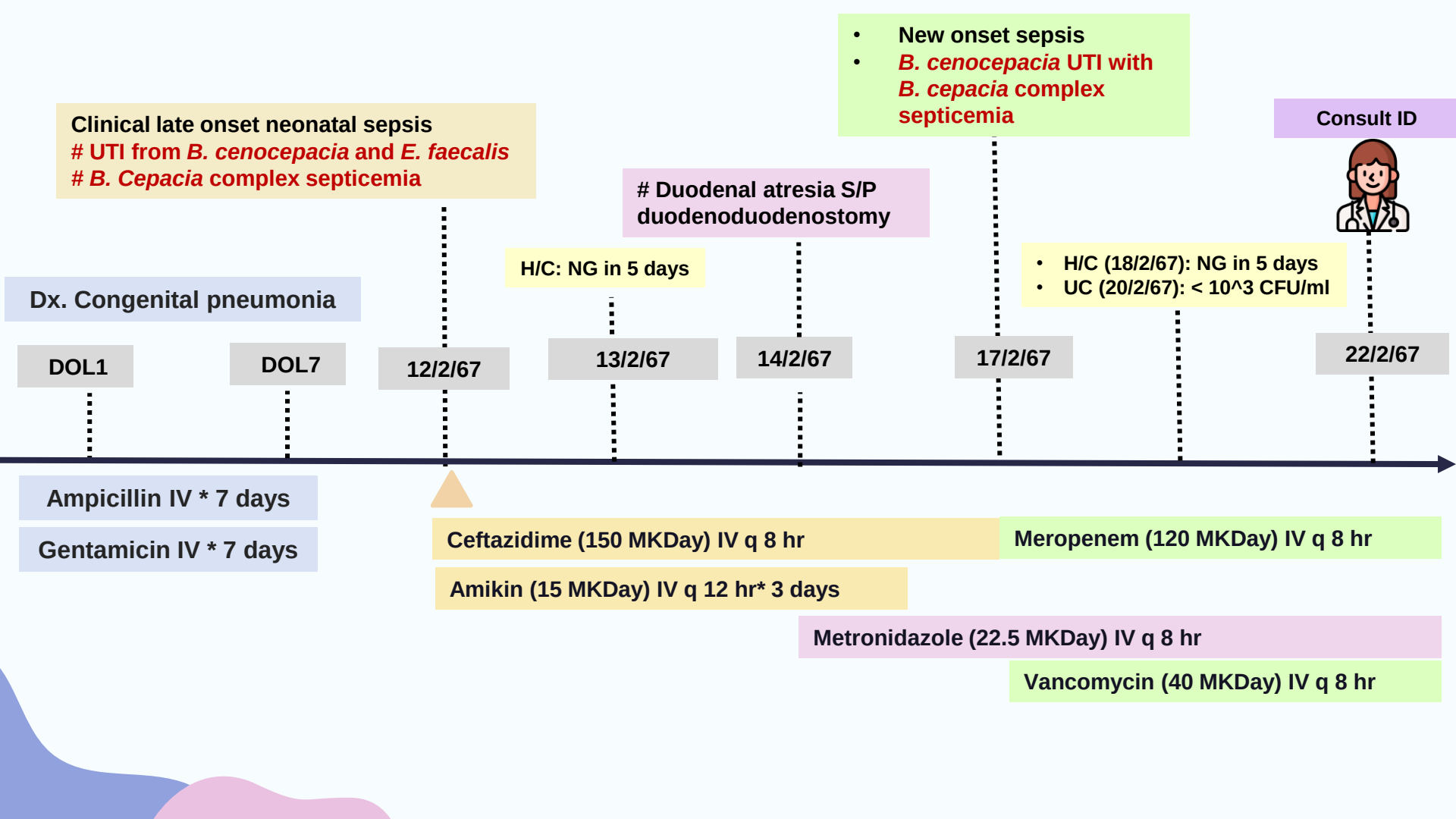
Request

Bottle2#
(แก้วที่ 2)

Comment

Preliminary report : gram negative bacilli

- Repeat H/C (18/2/67): pending





Progression

- **Impression:**
 - ***B. cenocepacia and E. faecalis UTI with B. cepacia complex septicemia***
- **Management:**
 - **Consult ped ID (22/2/67)**



Treatment: *B. cepacia* complex

- **Drugs that may have activity against *B. cepacia* complex include**
 - Trimethoprim-sulfamethoxazole
 - Ceftazidime
 - Minocycline
 - Fluoroquinolones
 - Carbapenems
 - Newer beta-lactam/beta-lactamase inhibitor combinations
 - Ceftazidime-avibactam or ceftolozane-tazobactam
- Some experts **recommend combinations of antimicrobial agents** that provide synergistic activity against *B. cepacia* complex in vitro
- **Intrinsically resistant to aminoglycosides, polymyxins, beta-lactam agents** (penicillin, ampicillin, carboxypenicillins, 1st and 2nd generation cephalosporins)



Management in this case

- ***B. cenocepacia* UTI with *B. cepacia* complex septicemia**
 - Combination therapy:
 - Continue meropenem (120 MKDay) IV q 8 hr
 - **Add TMP/SMX (10 MKDay of TMP) IV q 12 hr**
 - Repeat hemoculture
- ***E. faecalis* UTI**
 - Vancomycin (40 MKDay) IV q 6 hr for 7 days
 - Repeat urine C/S (20/2/67): $< 10^3$ CFU/ml





Repeat H/C and Urine C/S

- Repeat H/C (18/2/67): no growth in 5 days
- H/C (22/2/67, before start TMP/SMX): pending
- Repeat UA (20/2/67): WBC 1-2 HPF, RBC none, Epi none, Leukocyte esterase trace
- Urine G/S (20/2/67): no bacteria found
- Urine C/S (20/2/67) (single cath): $< 10^3$ CFU/ml



Other problems

Peripheral PS

- Heart: SEM at LUPSB Gr. II/VI radiate to back
- Consult cardio for echocardiogram
- Echocardiogram (22/2/67)
 - No structural heart defect (ASD, VSD, valvular PS, AS)
 - Turbulent flow at Left MPA Max gradient 16 mmHg
 - PFO 1 mm

Single umbilical artery

- U/S KUB (31/1/67): Small size of bilateral kidneys. The right and left kidney lengths are 3.4 cm and 3.4 cm, respectively. Normal echogenicity of bilateral renal parenchyma. No dilatation of bilateral pelvicalyceal systems

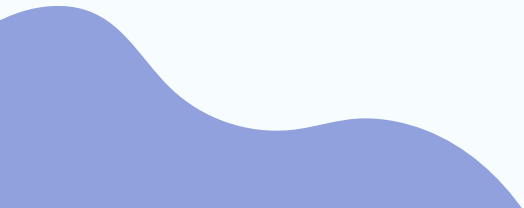
Neurogenic bladder: consult uroxx → suggest CIC q 4 hr

R/O syndromic

- Consult genetic → W/U Chromosome study: 46, XX



Burkholderia spp.





***Burkholderia* spp.**

- Gram-negative bacilli, non–lactose-fermenting
- ***B. cepacia* complex** comprises at least 22 species
- Other clinically important species of *Burkholderia* include
 - *Burkholderia pseudomallei*, *Burkholderia mallei*, *Burkholderia gladioli*, *Burkholderia thailandensis*, and *Burkholderia oklahomensis*.
- The incubation period for melioidosis (*B. pseudomallei*) is **1 to 21 days**, with a median of 9 days, but can be prolonged (years).



Clinical manifestations: *B. cepacia* complex

- Associated with infections in **cystic fibrosis (CF), chronic granulomatous disease (CGD), hemoglobinopathies, malignant neoplasm, preterm infants**
 - Cystic fibrosis → pneumonia, bronchiectasis
 - CGD → Pneumonia (most common), lymphadenitis
 - Disease onset: insidious → low grade fever early in the course and systemic effects occurring 3-4 weeks later
 - Pleural effusion
 - Lung abscess
- **Health care-associated infection** → wound, UTI, pneumonia
- **Clusters of disease**
 - Associated with contaminated pharmaceutical product
 - Nasal sprays, mouthwash, sublingual probes, prefilled saline flush syringes



Diagnostic tests: *B. cepacia* complex

- **Culture** is the appropriate method to diagnose *B. cepacia* complex infection
- Confirmation of identification of *B. cepacia* complex species by **mass spectrometry or by polymerase chain reaction** assay is recommended.



Waterborne Infectious Diseases in Health-Care Facilities

- *Legionellosis*
- Other Gram-Negative Bacterial Infections
 - *Pseudomonas aeruginosa*
 - *Burkholderia cepacia*
 - *Stenotrophomonas maltophilia*, *Sphingomonas* spp.
 - *Serratia marcescens*
 - *Acinetobacter* spp.
 - *Enterobacter* spp.

Table 13. Other gram-negative bacteria associated with water and moist environments

Bacteria	Implicated contaminated environmental vehicle
<i>Burkholderia cepacia</i>	- Distilled water ⁵²⁷ - Contaminated solutions and disinfectants ^{528, 529} - Dialysis machines ⁵²⁷ - Nebulizers ⁵³⁰⁻⁵³² - Water baths ⁵³³ - Intrinsically-contaminated mouthwash ⁵³⁴ (This report describes contamination occurring during manufacture prior to use by the health-care facility staff. All other entries reflect extrinsic sources of contamination.) - Ventilator temperature probes ⁵³⁵
<i>Stenotrophomonas maltophilia</i> , <i>Sphingomonas</i> spp.	- Distilled water ^{536, 537} - Contaminated solutions and disinfectants ⁵²⁹ - Dialysis machines ⁵²⁷ - Nebulizers ⁵³⁰⁻⁵³² - Water ⁵³⁸ - Ventilator temperature probes ⁵³⁹
<i>Ralstonia pickettii</i>	- Fentanyl solutions ⁵⁴⁰ - Chlorhexidine ⁵⁴¹ - Distilled water ⁵⁴¹ - Contaminated respiratory therapy solution ^{541, 542}
<i>Serratia marcescens</i>	- Potable water ⁵⁴³ - Contaminated antiseptics (i.e., benzalkonium chloride and chlorhexidine) ⁵⁴⁴⁻⁵⁴⁶ - Contaminated disinfectants (i.e., quaternary ammonium compounds and glutaraldehyde) ^{547, 548}
<i>Acinetobacter</i> spp.	- Medical equipment that collects moisture (e.g., mechanical ventilators, cool mist humidifiers, vaporizers, and mist tents) ⁵⁴⁹⁻⁵⁵⁶ - Room humidifiers ^{553, 555} - Environmental surfaces ⁵⁵⁷⁻⁵⁶⁴
<i>Enterobacter</i> spp.	- Humidifier water ⁵⁶⁵ - Intravenous fluids ⁵⁶⁶⁻⁵⁷⁸ - Unsterilized cotton swabs ⁵⁷³ - Ventilators ^{565, 569} - Rubber piping on a suctioning machine ^{565, 569} - Blood gas analyzers ⁵⁷⁰



Waterborne Infectious Diseases in Health-Care Facilities

Bacteria	Implicated contaminated environmental vehicle
<i>Burkholderia cepacia</i>	• Distilled water • Contaminated solutions and disinfectants • Dialysis machines • Nebulizers • Water baths • Intrinsically-contaminated mouthwash (This report describes contamination occurring during manufacture prior to use by the health-care facility staff. All other entries reflect extrinsic sources of contamination.) • Ventilator temperature probes



Measure to prevention

- Hand hygiene
- Glove use, barrier precaution
- Eliminating contaminated environment reservoirs



Point of learning

- Management of *B. cepacia* complex infection → Septicemia, UTI
 - Combination therapy → drug against activity to *B. cepacia*
- Infection control and prevention in health-care setting



Thank you

